

Deep Learning Enhanced Stereo Vision for LiDAR Emulation in Precise Robotic Arm Manipulation

Abstract—This paper presents a deep learning-augmented stereo vision system designed to emulate LiDAR-grade depth maps for precise robotic arm manipulation in static environments. We developed a prototype integrating stereo cameras with a Time-of-Flight (ToF) camera for ground-truth depth data and a custom 4-DOF robotic arm. A Pyramid Stereo Matching Network (PSMNet) was trained to refine stereo-derived point clouds by minimizing discrepancies with ToF data. In spite of difficulties from lighting variability and a limited training dataset, the custom-trained network significantly improved depth map accuracy, achieving a 27% reduction in End-Point Error (EPE) compared to a traditional SGBM baseline. These findings demonstrate that while deep learning-augmented stereo vision can approximate LiDAR-level mapping, its reliability is contingent on domain-specific training and controlled environmental conditions, positioning it as a viable, cost-effective solution for depth estimation in robotics.

Index Terms—stereo vision, deep learning, LiDAR emulation, robotic arm, depth estimation, PSMNet, camera calibration, SGBM

I. INTRODUCTION

Traditional methods for acquiring high-fidelity depth information frequently rely on Light Detection and Ranging (LiDAR) systems, recognized for their precision and reliability [1]. However, their high cost and complexity limit extensive utilization, particularly in static environments where cost-effective solutions are desirable [2].

Stereo vision systems present a viable and more economical alternative by reconstructing depth from visual discrepancies between images. While affordable, stereo vision encounters difficulties in depth accuracy, especially under varying lighting conditions and due to complex calibration requirements [3]. Recent progress in computer vision, particularly with deep learning, have profoundly influenced depth estimation [4]. The evolution from traditional geometric methods to deep learning approaches [5] indicates a trend towards data-driven methodologies to overcome inherent limitations. This research addresses this gap by investigating LiDAR emulation through an enhanced stereo vision system augmented with deep learning.

The primary objective is to achieve depth estimation accuracy comparable to LiDAR to facilitate precise robotic arm manipulation without high costs. This is realized by training Deep Neural Networks (DNNs) on datasets derived from

Time-of-Flight (ToF) cameras, which serve as a reliable source of ground truth depth data, to correct and refine stereo vision depth maps [6].

The key contributions of this study include:

- Development of a prototype system integrating a custom-built four-degree-of-freedom (4-DOF) robotic arm with a stereo camera setup and a Time-of-Flight (ToF) camera.
- Implementation of camera calibration and stereo rectification.
- Application of the Semi-Global Block Matching (SGBM) algorithm to establish a robust baseline for stereo depth estimation.
- Enhancement of stereo depth maps via a comparative deep learning analysis, evaluating the performance of pre-trained PSMNet models against a network custom-trained on our system's domain-specific ground truth data.
- Thorough examination of the system's effectiveness and limitations in achieving LiDAR-grade depth emulation.

This research frames the problem as an essential need for precise yet affordable depth sensing in robotics, positioning the deep learning-enhanced stereo vision as a viable solution.

II. RELATED WORK

The field of depth estimation has experienced substantial developments, particularly with the integration of deep learning techniques, which have addressed many traditional challenges associated with stereo matching. This section reviews recent contributions, contextualizing the presented approach within the broader academic landscape.

A. Deep Learning Advancements in Stereo Matching

Recent research has focused on enhancing stereo matching through novel deep learning architectures and training paradigms. Wen et al. [7] introduced FoundationStereo, a zero-shot stereo matching model that leverages large-scale synthetic datasets, demonstrating robust generalization.

Further developments integrate monocular depth cues into stereo frameworks; models like DEFOM-Stereo [8] and MonSter [9] leverage this hybrid approach to improve accuracy in challenging conditions such as occlusions and textureless regions.

Addressing the scarcity of real-world annotated stereo data, Wang et al. [10] developed StereoGen, a novel pipeline capable of generating high-quality stereo images from a single input using deep learning. This method synthesizes corresponding right images, enabling impressive generalization across various environments.

Comprehensive reviews, such as that by Rohan et al. [5], provide a systematic overview of deep learning-based depth estimation. These reviews highlight the evolution of techniques from monocular to stereo and multi-view approaches, emphasizing the critical role of large datasets and advanced neural architectures in improving depth estimation accuracy and generalization capabilities.

Such analyses underscore the ongoing challenges, including the need for extensive manual tuning and the difficulty of generalizing models to diverse, unseen scenes. While models pre-trained on large-scale public datasets (e.g., for autonomous driving or synthetic scenes) achieve state-of-the-art performance, their effectiveness can be limited by domain shift when applied to specialized tasks such as indoor robotic manipulation. This progression, coupled with the challenge of domain adaptation, directly informs our selection of a robust architecture like PSMNet and our methodological focus on evaluating the benefits of custom, domain-specific training.

B. Applications of Stereo Vision in Robotics

Stereo vision is a pivotal tool for various robotic applications, particularly those requiring precise 3D perception. Perezyabov et al. [11] demonstrated the utility of stereo cameras for road obstacle detection. Their method combines depth and image data to enhance detection accuracy, addressing the limitations of monocular systems.

In the context of robotic manipulation, depth perception plays a critical role in 3D object localization and interaction [12]–[15]. Many studies in this domain focus on 6D pose estimation for grasping, a task that relies heavily on accurate depth sensing [16]. The integration of advanced vision systems enables robots to perceive and interact with their environment with high precision, moving beyond simple pick-and-place operations to more complex, dexterous manipulations. The literature consistently highlights the critical need for accurate depth perception in robotic manipulation and the growing role of deep learning in enhancing vision systems for this purpose, reinforcing the practical relevance and timeliness of the current research.

III. METHODOLOGY

This section details the experimental setup, calibration procedures, and algorithms used to achieve deep learning-enhanced stereo vision.

A. Hardware Setup

The hardware consists of a custom robotic arm and an integrated vision system. The manipulation platform is a custom-built 4-DOF robotic arm fabricated from 3D-printed PLA, aluminium rods, actuated by NEMA 17 stepper motors

with TMC2208 drivers and controlled by an Arduino Uno with a CNC shield.

The vision system integrates a Time-of-Flight (ToF) camera (MaixSense-A075V) for ground-truth depth data [17] and two Fantech Luminous C30 cameras for stereo vision. The ToF camera has a depth resolution of 320x240 and an operating range of 0.15 to 1.5 meters. The stereo cameras provide a 2560x1440 resolution. The complete system, including the robotic arm and top-mounted cameras, is assembled within a structured workspace to ensure controlled data collection, as depicted in Figures 1, 2, and 3.

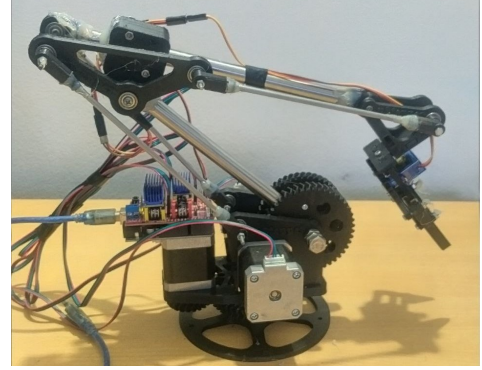


Fig. 1. Custom-built 4-DOF robotic arm.

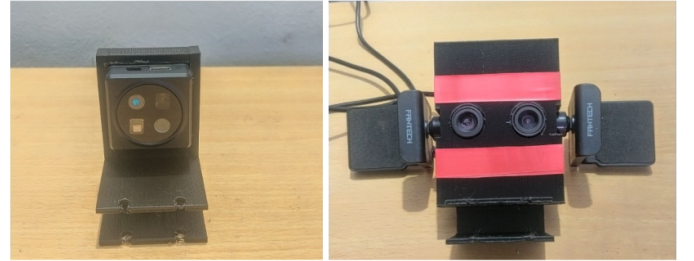


Fig. 2. ToF and Stereo Camera System.



Fig. 3. Workspace setup for data acquisition and robotic arm operation.

B. Camera Calibration

Camera calibration is a foundational step for geometric accuracy in 3D reconstruction [18]. The process corrects lens distortions and provides the parameters for aligning stereo images, which is critical for accurate depth estimation.

1) *Epipolar Geometry and Rectification*: Epipolar geometry describes the relationship between two camera views of a 3D scene. The camera's intrinsic matrix $\mathbf{K}$, shown in (1), defines its internal optical properties:

$$\mathbf{K} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix} \quad (1)$$

Extrinsic parameters ($\mathbf{R}$, $\mathbf{t}$) describe the camera's orientation and position relative to a world coordinate system, as defined in (2):

$$\mathbf{X}_c = \mathbf{R}\mathbf{X}_w + \mathbf{t} \quad (2)$$

Lens distortions are modeled using radial (K_1, K_2, K_3) and tangential (P_1, P_2) coefficients. The corrected image coordinates (x'', y'') are calculated from the distorted coordinates (x', y') using the model in (3) and (4):

$$x'' = x'(1 + K_1r^2 + K_2r^4 + K_3r^6) + 2P_1x'y' + P_2(r^2 + 2x'^2) \quad (3)$$

$$y'' = y'(1 + K_1r^2 + K_2r^4 + K_3r^6) + P_1(r^2 + 2y'^2) + 2P_2x'y' \quad (4)$$

where $r^2 = x'^2 + y'^2$. Since perfect hardware alignment is impractical, stereo rectification computationally transforms the images to align epipolar lines horizontally, simplifying the correspondence search for disparity calculation.

2) *Calibration Process and Results*: We performed calibration using a 9x7 checkerboard pattern, capturing images from multiple perspectives to estimate camera parameters with OpenCV [19], [20]. The derived intrinsic parameters and distortion coefficients are listed in Table I. The process resulted in a rectification **RMS error of 1.509**, confirming successful epipolar alignment as shown in the rectified view in Fig. 4.

TABLE I
FANTECH LUMINOUS C30 CAMERA INTRINSIC PARAMETERS AND DISTORTION COEFFICIENTS

Parameter	Value
f_x	680.9335952 mm
f_y	681.53275369 mm
c_x	322.60859724 px
c_y	257.99884455 px
K_1	$-3.84198386 \times 10^{-1}$
K_2	$3.37282195 \times 10^{-1}$
K_3	$1.72120651 \times 10^{-3}$
P_1	$-5.48946011 \times 10^{-4}$
P_2	$-6.45215020 \times 10^{-1}$

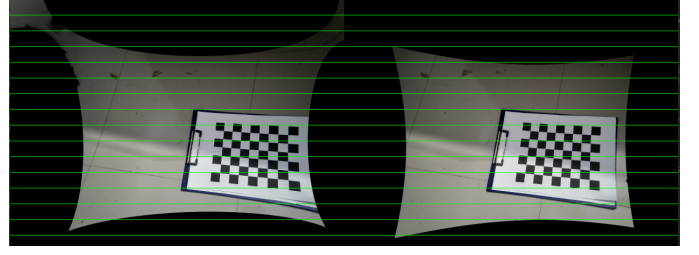


Fig. 4. Rectified camera view demonstrating epipolar alignment.

C. Data Acquisition

The data acquisition involved capturing synchronized stereo image pairs (I_L, I_R) and ToF depth maps across various scenes. Disparity $d(x, y)$, which represents the horizontal shift between corresponding points in the stereo pair, is defined in (5):

$$d(x, y) = x_L - x_R \quad (5)$$

Depth (Z) is then estimated from disparity via triangulation, following the fundamental stereo vision formula shown in (6):

$$Z = \frac{f \cdot B}{d(x, y)} \quad (6)$$

where f is the focal length and B is the baseline distance between the stereo cameras (84.85mm in our setup).

D. Depth Estimation (Traditional SGBM)

A baseline for depth estimation was established using the Semi-Global Block Matching (SGBM) algorithm [21], [22], chosen for its balance of accuracy and computational efficiency. The algorithm was configured with the empirically optimized parameters shown in Table II. The SGBM method produced reliable depth maps in well-textured regions but struggled with noise in low-texture areas (Fig. 5).

TABLE II
SGBM ALGORITHM OPTIMIZED PARAMETERS

Parameter	Value
Block Size	7 pixels
Disparity Range	128 levels
Minimum Disparity	5
P1 Penalty	$8 * 3 * 7^2$
P2 Penalty	$32 * 3 * 7^2$
Uniqueness Ratio	15
Speckle Window Size	100
Speckle Range	2
Pre-Filter Cap	63

E. Deep Learning Enhancement (PSMNet)

To overcome the limitations of SGBM, specifically in textureless regions, we employed the Pyramid Stereo Matching Network (PSMNet) architecture [23]. PSMNet leverages

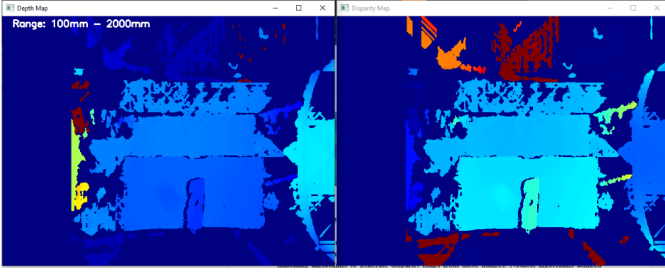


Fig. 5. Depth map and Disparity map generated using the SGBM algorithm.

global contextual information via a spatial pyramid pooling module and a 3D stacked hourglass module to regularize the matching cost volume, effectively resolving matching ambiguities (Fig. 6).

We conducted a comparative evaluation to quantify the performance gains. The End-Point Error (EPE) metric was used to compare the SGBM baseline against a pre-trained PSMNet (on SceneFlow and KITTI datasets) and a version of PSMNet fine-tuned for 1500 epochs on a custom-captured dataset from our experimental setup, using the ToF data as ground truth.

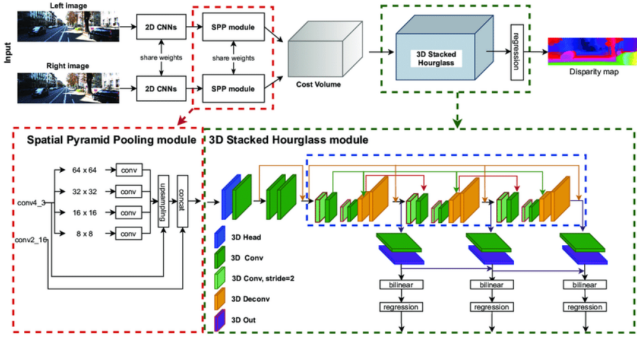


Fig. 6. Simplified architecture of the PSMNet [23]

IV. RESULTS

This section presents the comparative performance of the depth estimation models and discusses the implications of the findings.

The evaluation results, summarized in Fig. 7, confirm a clear performance hierarchy. The baseline SGBM algorithm yielded an End-Point Error (EPE) of 1.42. Pre-trained PSMNet models demonstrated improved performance, with the KITTI-trained model reaching an EPE of 1.18. Crucially, the custom-trained PSMNet achieved the lowest error (EPE=1.03), a **27% improvement** over the SGBM baseline.

Qualitative inspection of the generated 3D point clouds further supports these quantitative findings. The SGBM-derived point cloud (Fig. 8) exhibits significant noise and sparsity, particularly on uniform surfaces. In contrast, the point cloud from the custom-trained PSMNet (Fig. 9) is substantially denser and more coherent.

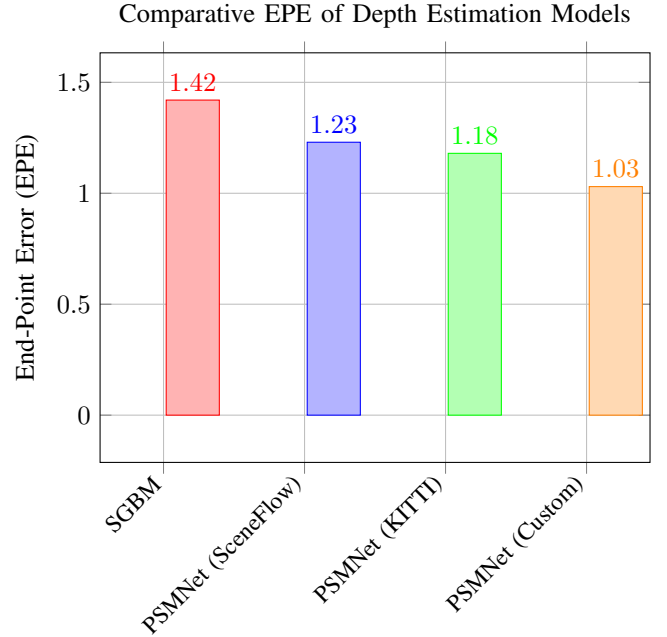


Fig. 7. Comparison of End-Point Error (EPE) for different depth estimation models against the ToF camera ground truth. The custom-trained PSMNet achieves the lowest error.

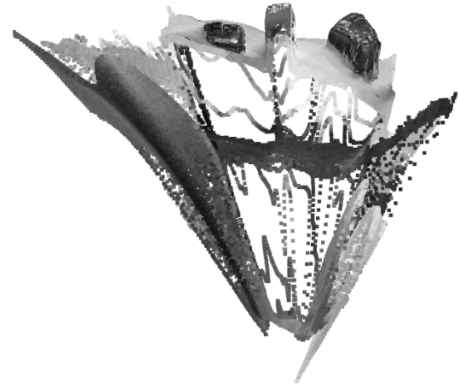


Fig. 8. Point Cloud generated from SGBM algorithm.

The superior performance of the custom-trained PSMNet model underscores the criticality of domain-specific fine-tuning for maximizing depth estimation accuracy. However, this quantitative success is tempered by several challenges. Despite the improvement, visual inspection of the PSMNet point cloud reveals residual artifacts and surface distortions, likely due to the limited size and diversity of our custom training dataset. Furthermore, the system remains sensitive to variations in ambient lighting, a persistent issue for passive vision sensors. These factors collectively indicate that while deep learning significantly enhances stereo vision, achieving LiDAR-grade reliability requires more extensive training data and robust solutions for environmental variability [24]. The current system can approximate LiDAR-level mapping, but its

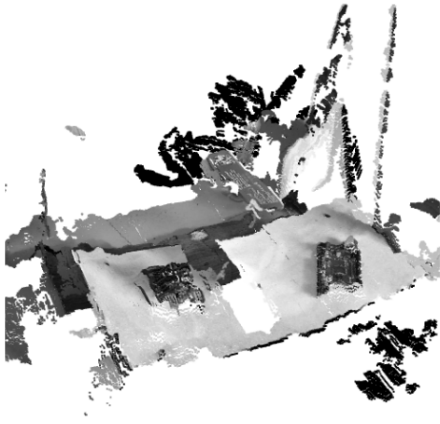


Fig. 9. Point clouds generated from the custom-trained PSMNet.

effectiveness is highly contingent on precise calibration and controlled conditions.

V. CONCLUSION

This research successfully demonstrated the potential of deep learning-enhanced stereo vision as a cost-effective alternative to LiDAR for high-precision robotic manipulation. Our principal finding is that a Pyramid Stereo Matching Network (PSMNet) fine-tuned on a domain-specific dataset significantly outperforms both traditional algorithms and pre-trained deep learning models. This custom-trained network achieved a 27% reduction in End-Point Error compared to the SGBM baseline, as well as a notable improvement over networks pre-trained on generic autonomous driving and synthetic datasets. This result underscores the critical impact of mitigating domain shift; while pre-trained models offer a strong foundation, adapting them to the specific sensor characteristics, lighting conditions, and object distributions of the target workspace is paramount for maximizing accuracy.

Consequently, our work validates this approach as a viable pathway toward affordable, high-fidelity 3D perception in controlled environments. It presents a clear trade-off for system designers: for applications where environmental conditions can be regulated, this method offers LiDAR-approximate performance at a fraction of the cost. Nevertheless, the system's performance remains contingent on the quality and scope of training data and is susceptible to ambient lighting variations, indicating that it has not yet achieved the all-conditions robustness of active sensors.

VI. FUTURE WORK

Building upon this work, our primary future direction is to enhance system robustness by addressing the current data limitations. We plan to create a larger and more diverse training dataset by integrating higher-fidelity sensors, such as a commercial LiDAR unit or an Intel RealSense camera. This will provide more precise ground-truth data and enable the model to generalize better across a wider range of objects and scenarios.

Second, to rigorously validate the practical utility of the enhanced depth maps, the current 4-DOF manipulator will be upgraded to a more rigid, industrial-grade 6-DOF platform. Incorporating force-torque sensing at the wrist will allow for quantitative evaluation of closed-loop manipulation tasks, such as peg-in-hole insertion or grasping objects with arbitrary poses, thereby directly correlating depth map accuracy with task success.

Further research will extend the system's application to dynamic environments. This involves investigating methods for real-time performance optimization and exploring multi-sensor fusion techniques that synergistically combine the dense, textured output of stereo vision with sparse but highly accurate data from other modalities. The goal is to create a perception system that is robust to environmental changes and moving objects. Finally, we will explore more advanced deep learning architectures, such as vision transformers or hybrid CNN-transformer models, which may offer improved handling of global context and long-range dependencies, potentially yielding further improvements in depth estimation accuracy and reducing artifacts in textureless regions.

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